MILITARY SPECIFICATION

PAPER, CHEMICAL AGENT DETECTOR: M9

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of paper detector which indicates the presence of liquid droplets of toxic chemical agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Box, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Box, Wood, Nailed and Lock Corner
PPP-B-636 - Box, Shipping, Fiberboard
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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MILITARY

MIL-P-116 - Preservation-Packaging: Methods Of
MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging
MIL-B-131 - Barrier Materials, Water Vaporproof, Flexible, Heat-Sealable
MIL-G-3056 - Gasoline, Automotive, Combat
MIL-P-51494 - Dye, B-1, Water Dispersed Formation

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-67-271 - Paper, Chemical Agent Detector, M9

PUBLICATIONS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

136-300-305 - Procedures for Aerosol Testing Liquid Agent Detector (LAD) Paper
136-300-306 - Procedures for Abrasion Testing Liquid Agent Detector (LAD) Paper

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

Munsell Book of Color, Matte Finish

(The Munsell Book of Color may be obtained from the Munsell Color Company, Inc., 2441 N. Calvert Street, Baltimore, MD 21218.)

Pantone Matching System 464U and 464C

(Pantone Matching System booklets may be obtained from Pantone, Inc., 55 Knickerbocker Road, Moonachie, NJ 07074.)

PRESSURE SENSITIVE TAPE COUNCIL (PSTC)

PSTC-1 - Peel Adhesion For Single Coated Tapes 180° Angle

(Application for copies should be addressed to the Pressure Sensitive Tape Council, 1201 Waukegan Road, Glenview, IL 60025.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T400 - Sampling a Single Lot of Paper, Paperboard or Related Products
T404 - Tensile Breaking Strength of Paper and Paperboard
T410 - Weight Per Unit Area (Basis Weight or Substance) of Paper or Paperboard
T411 - Thickness (Caliper) of Paper and Paperboard
T412 - Moisture in Paper
T441 - Water Absorptiveness of Sized (Non-Bibulous) Paper and Paperboard (Cobb Test)
T494 - Tensile Breaking Properties of Paper and Paperboard (Using Constant Rate of Elongation Apparatus)
T614 - Analysis of Alum
T621 - Analysis of Rosin

(Applications for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, NY 10017.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. The M9 paper shall consist of a roll of chemical agent detector paper assembled from B-1 paper (3.1.1) and backing (3.1.2), contained in a dispenser box conforming to Drawing 5-67-272.

3.1.1 B-1 paper.

3.1.1.1 Materials. B-1 paper shall be manufactured from the following materials (see 6.3):

(a) Paper pulp. The paper pulp shall consist of 80 percent bleached softwood sulfate and 20 percent bleached hardwood sulfate.

(b) Water dispersed formation of B-1 dye. The water dispersed formation of B-1 dye shall be material conforming to MIL-D-51494, which has been dispersed in water using a colloid mill (see 6.4), and then passed through a No. 325 sieve prior to mixing with the paper pulp. A written certification that the B-1 dye meets the particle size requirements of table I, MIL-D-51494, shall be required prior to its use in the B-1 paper.

(c) Alum. The alum shall be a basic, iron-free papermaker's alum as defined in TAPPI T614.

(d) Rosin. The rosin shall be a natural rosin as defined in TAPPI T621.

(e) Resin. The resin shall be a melamine formaldehyde resin - American Cyanamid's "Parez 607," or equivalent.

(f) Camouflage colors. A combination of pigment blue 15 (Color Index No. 74160), yellow iron oxide, and calcatone black NI paste (American Cyanamid) shall be used as camouflage colors, in conjunction with the required amount of B-1 dye (see table I), to meet the color requirements of table IV.

3.1.1.2 Chemical and physical characteristics. The B-1 paper shall conform to the chemical and physical characteristics of table I when tested as specified therein.

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TABLE I. Chemical and physical characteristics of B-1 paper

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Basis weight, grams per square meter	80.0	86.0	4.4.4.1.1
Dry tensile, machine direction, kilograms per 15-mm (MD)	6.0		4.4.4.1.2
Caliper (thickness), millimeters	0.140		4.4.4.1.3
Moisture, percent by weight	4.5	7.5	4.4.4.1.4
B-1 dye, percent by weight	0.7	0.9	4.4.4.1.5
Sizing (Cobb) test (felt and wire side), grams per square meter	18	24	4.4.4.1.6

3.1.2 Backing.

3.1.2.1 Materials. The backing shall be manufactured from the following materials:

(a) Mylar. The mylar shall be a matte mylar, DuPont 92EB11 or smooth mylar, DuPont 92S or equal.

(b) Adhesive. The adhesive shall be a pressure-sensitive adhesive, H.B. Fuller SC1204. No more than 6 months shall have passed from the date of manufacture of the adhesive to the date of manufacture of the backing. The adhesive shall be in one phase (no separations) and shall be free of particulate matter.

(c) Release liner. The release liner shall be two-side silicone coated release liner, Akrosil Paper Company NS 62UBL10-2 or equal. The release liner shall be a brown color no lighter than Pantone Matching System (PMS) 464U or 464C or equal. No more than 6 months shall have passed from the date of manufacture of the release liner to the date of manufacture of the backing.

3.1.2.2 Manufacture. The backing shall consist of a mylar strip coated on both sides with adhesive. One of the sides shall be covered with a release liner and the entire item shall be rolled with the exposed adhesive side on the inside of the roll (see figure 1). For the B layer, a 1/8 $\pm$ 1/16 inch border containing no adhesive on either edge of the backing material is permitted.

3.1.2.3 Physical characteristics. The backing shall conform to the physical characteristics of table II when tested as specified therein.

3.1.2.4 Workmanship. The backing shall be rolled with the edges perpendicular to the core and with the release liner on the outside of the roll. The edges of the roll shall be smooth and free from adhesive. The adhesive coatings on the mylar shall be uniform and free from excess residual solvent.

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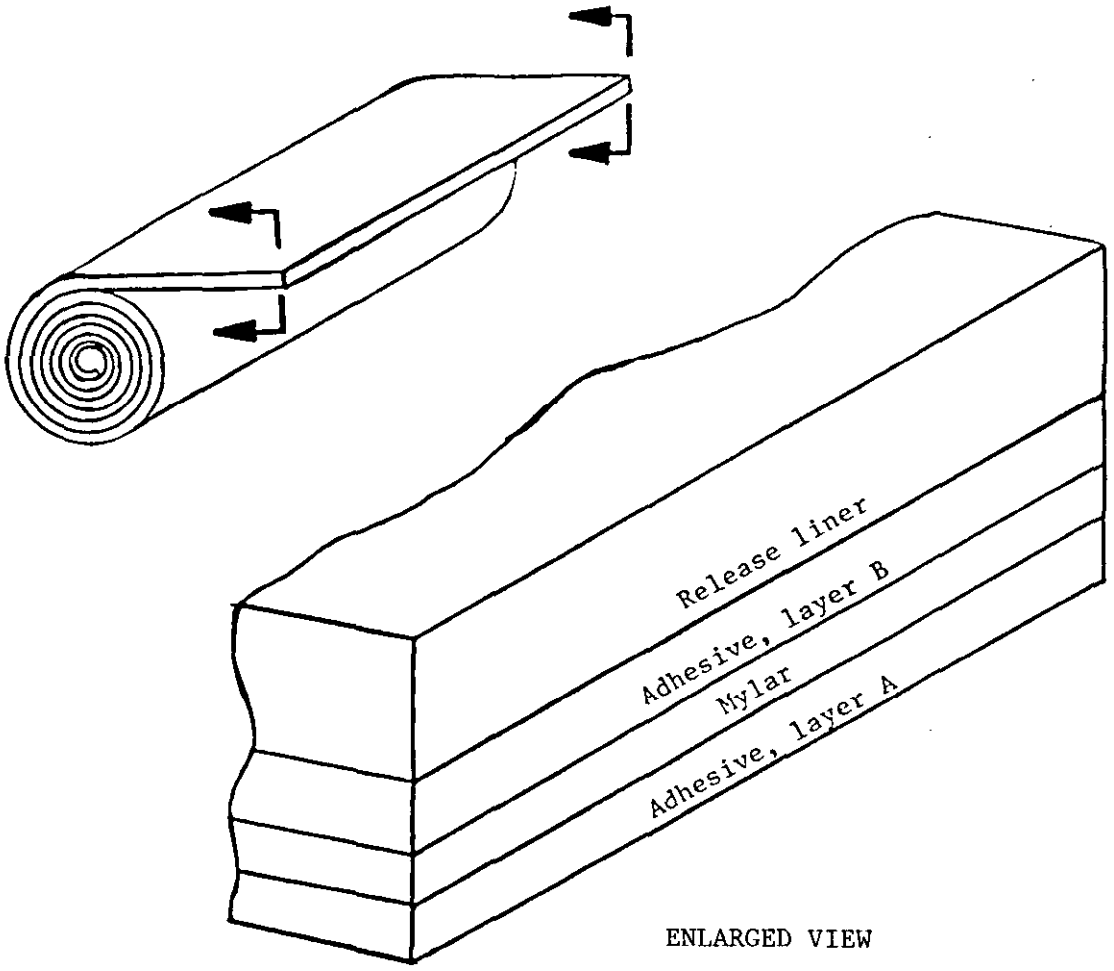


FIGURE 1. Backing.

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TABLE II. Physical characteristics of backing

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Backing thickness, mils	7.0	8.1	4.4.4.2.1
Adhesive thickness, dry, mils:			
A-layer	0.9	1.1	4.4.4.2.1
B-layer	1.7	1.9	4.4.4.2.1
Mylar thickness, mils	0.9	1.1	4.4.4.2.1
Release liner thickness, mils	3.5	4.0	4.4.4.2.1

3.2 Manufacture and assembly. The M9 paper shall be manufactured by laminating B-1 paper to the backing in accordance with Drawing 5-67-269, and assembled in accordance with Drawing 5-67-271.

3.3 Physical characteristics. The M9 paper shall conform to the physical characteristics of table III when tested as specified therein.

TABLE III. Physical characteristics of M9 paper

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Roll dimensions:			4.4.4.3.1
Length, feet	30.00	30.75	
Width, inches	1.97	2.00	
Peel strength from steel, oz per inch	80	128	4.4.4.3.5

3.4 Volatile matter. Packaged M9 paper shall contain no more than 2.0 percent volatile matter when tested as specified in 4.4.4.3.3.

3.5 Tearability. The rolled detector paper shall pull easily from the dispenser and tear with clean separation across the serrated edge when tested as specified in 4.4.4.3.1.

3.6 Color. The color of the paper shall fall within the range of colors defined by the Munsell colors listed in table IV when tested as specified in 4.4.4.3.2.

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TABLE IV. Munsell colors for M9 paper

Hue	Value	Chroma
2.5Y	6	1,2
5Y	6	1,2
7.5Y	6	1,2
10Y	6	1,2
2.5GY	6	1,2
5GY	6	1,2

3.7 Workmanship. The roll of detector paper shall be uniform in color when viewed from a distance of 5 feet, with no specks, spots, or light or dark patches, and shall be free from rough or uneven edges, holes, cuts, tears, thin spots, wrinkles, grease, oil, dirt, and other foreign matter. Tunneling or creasing shall not be severe enough to cause scuffing or problems in dispensing the laminated paper from the dispenser box. The dispenser box shall be free from damage and grease, oil, dirt, and other foreign matter. (See 4.4.4.3.1.)

3.8 Preproduction. Prior to the start of regular production, a preproduction sample of M9 paper shall be produced in accordance with this specification for examination and test (see 4.3).

3.8.1 Functioning. The M9 paper, in the presence of chemical agent or simulant as specified (see 6.2), shall change to a color in the range of pink-red to red, and the color shall develop in no more than 20 seconds when tested as specified in 4.3.4.1.

3.8.2 Interferences. The M9 paper shall not be affected by water or gasoline when tested as specified in 4.3.4.2.

3.8.3 Storage resistance. The M9 paper shall not show a pink coloring or any spots after storage at 90°C for 12 days and shall meet the functioning requirements of 3.8.1 when tested as specified in 4.3.4.3.

3.8.4 Scuff resistance. The M9 paper shall not show red scuff marks when tested as specified in 4.3.4.4.

3.9 Release liner adhesion, M9 paper. When tested as specified in 4.4.4.3.1, the adhes ive shall adhere to mylar surfaces only. No adhesive deposit or residue on release liner shall occur.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is respon sible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government's responsibility. The Government is responsible for the performance of the preproduction tests specified in 4.3.4.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of 500 M9 papers shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production.

4.3.2 Sampling. See 6.7 for sampling and testing precautions.

4.3.2.1 For examination.

(a) Backing. Each roll of backing shall be examined.

(b) M9 paper. Sampling shall be conducted in accordance with MIL-STD-105. The sample unit shall be one unit pack.

4.3.2.2 For test.

(a) B-1 paper. A 2-foot sample shall be selected from one end of each production roll. The 2-foot sample shall extend across the full width of the roll.

(b) Backing. Sampling for test shall be as specified in table V.

(c) M9 paper. Fifty M9 paper unit packs shall be randomly taken for test.

4.3.3 Inspection procedure.

4.3.3.1 For examination.

(a) Backing. Inspection procedure for examination shall be as specified in 4.4.3.1.1.

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(b) M9 paper. Each unit pack shall be examined in accordance with 4.4.3.1.2(b) for defects 101, 102, 103, 104, 105, 107, 108, and 109. Each dispenser contained therein shall be examined in accordance with the classification of defects in 4.4.3.1.2(a).

4.3.3.2 For test.

(a) B-1 paper. The 2-foot sample shall be tested as specified in 4.4.4.1. Failure of any test shall be cause for rejection of the roll of B-1 paper represented.

(b) Backing. Inspection procedure shall be as specified in table V. Prior to testing as specified in 4.4.4.2, each sample shall be kept at standard conditions of 23° ± 2°C and 50 ± 2 percent relative humidity for 24 hours.

(c) M9 paper. Twenty M9 paper unit packs shall be tested as specified in 4.3.4.3. Another 20 M9 paper unit packs shall be tested for leakage as specified in 4.4.4.3.4. After these 20 M9 paper unit packs have been tested for leakage, 10 shall be tested as specified in 4.3.4.4, 4.4.4.3.1, and 4.4.4.3.2, and eight shall be tested as specified in 4.4.4.3.5. The remaining 10 untested M9 paper unit packs shall be tested as specified in 4.4.4.3.3.

4.3.4 Tests. See 6.7 for sampling and testing precautions. Tests shall be conducted as follows:

4.3.4.1 Functioning. Determine conformance to functioning requirements (3.8.1) in accordance with Test Procedure 136-300-305.

4.3.4.1.1 Apparatus.

(a) Aerosol generator
(b) Spray needles

4.3.4.1.2 Procedure.

(a) Using individual specimens of M9 paper, generate:

(1) 100 ± 5 micrometer diameter droplets of tris-butoxylethylphosphate.
(2) 100 ± 5 micrometer diameter droplets of diacetone alcohol.
(3) 100 ± 5 micrometer diameter droplets of VX.
(4) 110 ± 5 micrometer diameter droplets of HD.
(5) 110 ± 5 micrometer diameter droplets of GD.

(b) For each of (a) above, record:

(1) Color change
(2) Time for color change

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4.3.4.2 Interferences. Use the same apparatus as in 4.3.4.1.

4.3.4.2.1 **Water.** Generate 1000 ± 50 micrometer diameter droplets of water on a specimen of M9 paper. After drying there shall be no visible mark or effect on the specimen of M9 paper.

4.3.4.2.2 Gasoline (conforming to MIL-G-3056).

(a) Generate 600 ± 30 micrometer diameter droplets of gasoline on a specimen of M9 paper. No color change shall develop within 10 seconds.

(b) Generate 1000 ± 50 micrometer diameter droplets of gasoline on a specimen of M9 paper. Any color developed within 10 seconds shall be no more than a faint pink blush.

4.3.4.3 Storage resistance.

4.3.4.3.1 Item preparation. Divide the sample in half, 10 for test and 10 for control. Identify by a suitable means each test item and each control item. Place the test items in a clean, covered container, in a clean, vapor-free oven for 12 days, maintained at 90° ± 2°C. Retain the control items at 25° ± 2°C for the 12 days.

4.3.4.3.2 Procedure. Remove the test items from the oven and allow to cool to room temperature. Visually inspect the heat seal on the unit pack to insure that it has not opened during storage. Inspect the test items for conformance to 3.8.3 using the control items as standards. Determine conformance to functioning (3.8.1) by testing 25 samples from the test items (three from each of five test items and two from the remaining five test items) and 25 samples from the control items (three from each of five control items and two from the remaining five control items) in accordance with 4.3.4.1. Retain sufficient control items and test them for interference (4.3.4.2).

4.3.4.4 Scuff resistance. Determine the scuff resistance of the paper in accordance with Test Procedure 136-300-306.

4.4 Quality conformance inspection.

4.4.1 Lotting.

4.4.1.1 B-1 paper. A lot shall consist of the production of rolls of B-1 paper, produced by one manufacturer at one time, from the same materials, and under essentially the same manufacturing conditions. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.4.1.2 Backing. A lot shall consist of the rolls of backing, produced by one manufacturer at one time, from the same materials, and under essentially the same manufacturing conditions. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5). A lot of backing shall not contain more than one lot of adhesive.

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4.4.1.3 M9 paper. A lot shall consist of the M9 paper produced by one manufacturer at one time, from the same materials, and under essentially the same manufacturing conditions. A lot shall contain not more than one lot of backing material or one lot of B-1 paper.

4.4.2 Sampling. See 6.7 for sampling and testing precautions.

4.4.2.1 For examination.

(a) Backing. Each roll of backing shall be examined.

(b) M9 paper. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test.

(a) B-1 paper. A 2-foot sample shall be selected from one end of each production roll. The 2-foot sample shall extend across the full width of the roll. In addition, enough B-1 paper shall be selected to make 100 M9 papers.

(b) Backing. Sampling for test shall be as specified in table V.

(c) M9 paper. Sampling shall be conducted prior to unit packing in accordance with MIL-STD-105, level S-2. The sample unit shall be one roll of M9 paper.

TABLE V. Sampling and inspection procedure for backing tests

Requirement		Sample	Test	Acceptance/rejection
Characteristic	Paragraph	size	paragraph	criteria (for roll of backing)
Thickness:	3.1.2.3	3 specimens from one end of each roll	4.4.4.2.1	
Backing				Accept 1, reject 2
Adhesive (each coating)				Accept 1, reject 2
Mylar				Accept 1, reject 2
Release liner				Accept 1, reject 2

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4.4.3 Inspection procedure. See 6.7 for sampling and testing precautions.

4.4.3.1 For examination.

4.4.3.1.1 Backing. A four-foot sample shall be taken from one end of each roll for examination across the full width of the roll. More than two workmanship defects for any inspection sample shall be cause for rejection of the roll. Defects are classified as follows:

(a) Edge of roll irregular or not perpendicular
(b) Excess adhesive
(c) Excess solvent

4.4.3.1.2 M9 paper. Dispensers and packaging shall be examined in accordance with MIL-STD-105 and the following classification of defects:

(a) Dispenser.

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dispenser (5-67-272) not as specified
102	Label instructions (5-67-273 and 5-67-274) illegible, missing, or not as specified
103	Marking (5-67-275) illegible, missing, or not as specified
104	Detector paper not visible
105	Protector disc not as specified

(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Paperboard protection (5-67-270) over metallic edge of box missing or inadequate
102	Polyethylene bag (PL5-67-271, item 6) missing or damaged
103	Contents not as specified
104	Barrier bag damaged
105	Notch or "TEAR HERE, ETC" marking missing from barrier bag
106	Quantity per shipping container not as specified
107	Container (unit or shipping) not as specified
108	Container closure not as specified

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109	Marking missing, incomplete, illegible, or not durable on unit or shipping container or palletized load
110	Preservation on level A pack missing or not dry
111	Shipping containers not palletized as specified

4.4.3.2 For test.

(a) B-1 paper. The 2-foot sample shall be tested as specified in 4.4.4.1.1 through 4.4.4.1.6. Failure of any test shall be cause for rejection of the roll of B-1 paper represented. The additional B-1 paper shall be tested as specified in 4.4.4.1.7. Failure of any manufactured M9 papers to meet the storage resistance test shall be cause for rejection of the lot of B-1 paper represented.

(b) Backing. Inspection procedure shall be as specified in table V. Prior to testing as specified in 4.4.4.2, each sample shall be kept at standard conditions of 23° ± 2°C and 50 ± 2 percent relative humidity for 24 hours.

(c) M9 paper. Each M9 paper sample shall be tested as specified in 4.4.4.3. Failure of any test by any sample shall be cause for rejection of the lot represented.

4.4.4 Tests. See 6.7 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.4.4.1 B-1 paper.

4.4.4.1.1 Basis weight. Determine basis weight in accordance with TAPPI T410.

4.4.4.1.2 Tensile strength. Determine tensile strength in machine direction in accordance with TAPPI T404 or T494; use 15-millimeter (mm) wide sample strips.

4.4.4.1.3 Caliper. Determine caliper in accordance with TAPPI T411.

4.4.4.1.4 Moisture. Determine moisture in accordance with TAPPI T412.

4.4.4.1.5 B-1 dye.

(a) Preparation of calibration curve. Weigh 1.00 gram (g) of standard dye sample (see 6.6) to the nearest 0.1 milligram (mg) and dissolve in acetone in a 500-milliliter (mL) volumetric flask at 25°F. Make up to volume and mix well (Solution A). From Solution A, prepare 4 additional solutions. Pipet 40-, 30-, 20-, and 10-mL aliquots of Solution A into separate 100-mL volumetric flasks and make each up to volume with acetone at 25°C (Solutions B, C,

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D, and E, respectively). Determine absorbance of Solutions A, B, C, D, and E with a suitable spectrophotometer at 510 nanometers (nm) using acetone as a blank. Plot the absorbance versus concentration (grams of B-1 dye per 100 mL of solution) on graph paper.

(b) Procedure. Weigh approximately 12.5 square centimeters of B-1 paper to the nearest 0.1 mg and cut it into small pieces (approximately 1/4- by 1/4-centimeter) into a beaker. Add 5 to 10 mL of acetone to the beaker and swirl to extract the B-1 dye. Decant the liquid into a 100-mL volumetric flask. Repeat the extraction (addition of acetone, swirling, and decanting) until the extraction is colorless, then make two further extractions. Make up to volume with acetone at 25°C and mix well. Determine absorbance at 510 nm. From the calibration curve, determine B-1 dye concentration for the corresponding absorbance. Calculate percent B-1 dye in the paper as follows:

$$\text{Percent B-1 dye} = \frac{100\ C}{W}$$

where: C = Concentration of B-1 dye in extract, grams per 100 mL, and
W = Weight, in grams, of paper sample.

4.4.4.1.6 Sizing test. Determine paper sizing characteristics in accordance with TAPPI T441 using 120 seconds exposure time.

4.4.4.1.7 Storage resistance of M9 paper manufactured using B-1 paper. Make 100 M9 papers using the same methods, materials, equipment, and processes of manufacture as will be used during regular production. Test 20 of these M9 papers in accordance with 4.3.4.3.

4.4.4.2 Backing.

4.4.4.2.1 Thickness.

(a) Apparatus. Determine all specified thickness measurements with a standard thickness gauge of the electronic variety such as the General Electric Thickness Gauge, Type B, or the Twin City Permascope, or equal.

(b) Procedure.

(1) Determine total thickness of the backing (release liner, B-layer of adhesive, mylar, and A-layer of adhesive). Record as T.

(2) Prepare a 50-50 mixture of chloroform and toluene and carefully wash off the A-layer of adhesive. Blot dry with an absorbent paper such as filter paper, and measure its thickness (release liner, B-layer of adhesive, and mylar). Record as X.

(3) Remove the release liner. Wipe the release liner with a swab of solvent mixture, blot dry, and measure its thickness. Record as R.

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(4) Wash off the B-layer of adhesive from the mylar using the solvent mixture, blot dry, and measure its thickness. Record as M.

(c) Calculations.

$$A = T - X$$
$$B = X - (R + M)$$
$$T = R + B + M + A$$

where: T = Thickness of backing, in mils,
X = Thickness of release liner, B-layer of adhesive, and mylar, in mils,
A = Thickness of A-layer of adhesive, in mils,
R = Thickness of release liner, in mils,
M = Thickness of mylar, in mils, and
B = Thickness of B-layer of adhesive, in mils.

4.4.4.3 M9 paper.

4.4.4.3.1 Dimensions, tearability, and workmanship. Pull the paper from dispenser and tear across the cutter in approximately 3-foot lengths until the dispenser is empty.

(a) Record the length of each piece to the nearest 1/16 inch. Total the lengths and check for conformance to table III.

(b) Randomly measure the width of each length for conformance to table III.

(c) While dispensing the paper, determine conformance to 3.5.

(d) While dispensing the paper, determine conformance to 3.7.

(e) While dispensing the paper, determine conformance to 3.9.

4.4.4.3.2 Color. Determine the color of the paper by comparison with the Munsell Book of Color.

4.4.4.3.3 Volatile matter. Cut and discard the waterproof bag, plastic bag, and rubber band. Weigh the dispenser box with M9 paper and place it in an oven at 100° to 104°C for 48 hours. Remove the item from the oven, place it in a desiccator, allow it to cool for 15 minutes, and then reweigh it. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (A - B)}{A}$$

where: A = Weight of dispenser box with M9 paper before drying, and
B = Weight of dispenser box with M9 paper after drying.

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4.4.4.3.4 Leakage. Submerge the unit pack for at least 60 seconds in water heated to 58.5° ± 1.5°C. Keep the unit pack submerged in a vertical position with the heat sealed seam uppermost and covered by no more than 25 mm of water. Continuously examine the unit pack during the time it is submerged for evidence of bubbles. The presence of bubbles shall be considered evidence of leakage.

4.4.4.3.5 Peel strength, M9 paper. Determine peel strength in accordance with PSTC-1 with the following exceptions:

(a) Each prepared steel panel, with the M9 paper attached, is conditioned for 24 hours at 23° ± 2°C and 50 ± 2 percent relative humidity prior to test, and

(b) Paragraph 5.3 of PSTC-1 does not apply.

Take one sample from each of eight rolls of M9 paper. More than two failures shall be cause for rejection.

5. PACKAGING

5.1 Unit packing, level A. One roll of paper shall be packaged as specified on Drawing 5-67-271, in a heat sealable bag in accordance with method IA8 of MIL-P-116. The heat-sealable bag shall have inside dimensions of 7-3/4 by 7 (+1/4 or -0) inches; shall conform to type I, class E of MIL-B-117; and shall be formed from spun-bonded polyolefin backed material conforming to class 1 of MIL-B-131. The paperboard protector shall be folded over the serrated cutting edge of the metal edge stays, preventing contact of the metal with the heat-sealable bag. The polyethylene bag, item 6, Drawing PL 5-67-271, shall be neatly folded parallel to the resealable edge before inserting it into the heat-sealable bag. The heat-sealable bag shall be "nicked" and marked with the instruction "TEAR HERE TO OPEN" as specified for class E bags in MIL-B-117. Excess air shall be evacuated from the bag to the extent needed to assure that the folded bag conforms to the contours of the dispenser prior to heat sealing. The unit pack shall show no evidence of leakage when tested as specified in 4.4.4.3.4.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. One hundred unit packs, prepared as specified in 5.1, shall be packed upright in four layers of five rows of five unit packs, in a wooden box conforming to type overseas, style A, B, or I, grade A of PPP-B-601; class 2, style 4, grade A of PPP-B-621; or class 3, style optional of PPP-B-585. Each box shall be for a type 2 load of less than 100 pounds of contents, and shall have inside dimensions of 17-3/8 by 13 by 14 inches length, width, and depth, respectively. The wooden components of the PPP-B-585 box shall be preserved in the manner as specified for the PPP-B-601 box. All inside faces of the box shall be furnished with full face sized pads of fiberboard conforming to grade W6c of PPP-F-320. The motion of contents shall be prevented by inserting additional pads, as needed. Each box shall be closed and reinforced as specified in the applicable box specification.

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5.2.2 Level B. One hundred unit packs, prepared as specified in 5.1, shall be packed level B in the same manner as level A except that the box shall conform to grade V3c of PPP-B-636.

5.2.3 Commercial. A specified quantity (see 6.2) of unit packs prepared as specified in 5.1 shall be packed commercially as specified in MIL-STD-1188.

5.3 Unitization. Level A and level B packs shall be palletized as shown on Drawing 5-67-278. When specified (see 6.2), commercial packs shall be palletized in a manner to assure carrier acceptance and safe delivery at first destination.

5.4 Marking. Each level A unit pack shall be marked in accordance with MIL-STD-129 showing lot number, date of manufacture of item, and discard date. Level A and B packs shall be marked as shown on Drawing 5-67-277. Pallet loads of level A and B packs shall be marked as shown on Drawing 5-67-278. Commercial packs shall be marked in accordance with MIL-STD-1188; pallet loads of commercial packs shall be marked to assure ease of identification of the nature of the load and its origin and destination.

6. NOTES

6.1 Intended use. M9 detector paper is intended to be affixed to personnel or vehicles to indicate the presence of liquid droplets of toxic chemical agents in sufficient time for protective action to be taken.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Whether agent, simulant, or both shall be used for color change test (see 3.8.1.1)
(c) Level of packing required (see 5.2)
(d) For commercial packing, number of unit packs per commercial pack (see 5.2.3)
(e) When commercial packs shall be palletized (see 5.3)

6.3 Manufacturing data. Satisfactory B-1 paper has been made at Chemical Systems Laboratory on a pilot size (12-inch) fourdrinier using a mixture of 20 percent bleached hardwood sulfate and 80 percent bleached softwood sulfate. The materials listed in 3.1.1.1 were added to the paper slurry based on the oven dried weight of pulp in the following proportions:

Alum: 3 percent
Parez: 3 percent
Rosin: 2 percent
Pigment blue 15: 0.16 percent
Yellow iron oxide: 1.60 percent
Black pigment: 0.05 percent*

*Note: Black pigment used was a slurry containing 35 percent solids.

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A retention aid may be used to reduce the loss of dye fines. Satisfactory results have been obtained using 0.1 percent Nalco 623 SC.

6.4 Dispersion/colloid mill. Satisfactory results have been obtained using a Kady Dispersion Mill.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Standard dye sample. A standard sample may be obtained from Commander, US Army Armament Research and Development Command, ATTN: DRDAR-CLC-E, Aberdeen Proving Ground, MD 21010.

6.7 Sampling and testing precautions. This specification covers inspection of chemical material (B-1 dye) which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

6.8 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Descriptions EA-P-1162, dated 13 June 1980, EA-P-1163, dated 13 June 1980, and EA-B-1165, dated 13 June 1980.

Custodian:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51493(EA)	PAPER, CHEMICAL AGENT DETECTOR: M9

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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